Visualization Draft

Phoebe Dupa, Rebecca Martinez, and Shucan Zhao

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1. Set up

Packages

Data

2. Clean data

Cleaning weather data

Goal

Clean the weather dataset so it includes date, precipitation, maximum temperature, minimum temperature, month, year, and water year.

Cleaning metadata

Goal

Clean the metadata dataset so it includes survey date, site information, water year, and readable site names.

Cleaning water parameter data

Goal

Clean the water quality parameter dataset so it includes elevation code, elevation labels, dissolved oxygen, temperature, salinity, and conductivity.

3. Join data frames

Water quality = metadata + water parameters

Water quality weather = water quality + weather

4. Filter data for dissolved oxygen analysis

Goal

Create a filtered dissolved oxygen dataset that focuses on the three main NCOS sites, the 2024 and 2025 water years, and the main elevation codes used in the class example.

5. Summarize dissolved oxygen

Dissolved oxygen by site

```
# A tibble: 3 x 6
  site_full      mean_do median_do sd_do var_do     n
  <chr>            <dbl>     <dbl> <dbl>  <dbl> <int>
1 East Channel      7.22      7.28  3.43  11.8    109
2 Phelps Bridge     4.30      3.7   2.51   6.33   166
3 Venoco Bridge     5.68      5.98  4.01  16.1    257
```

Dissolved oxygen by site and elevation code

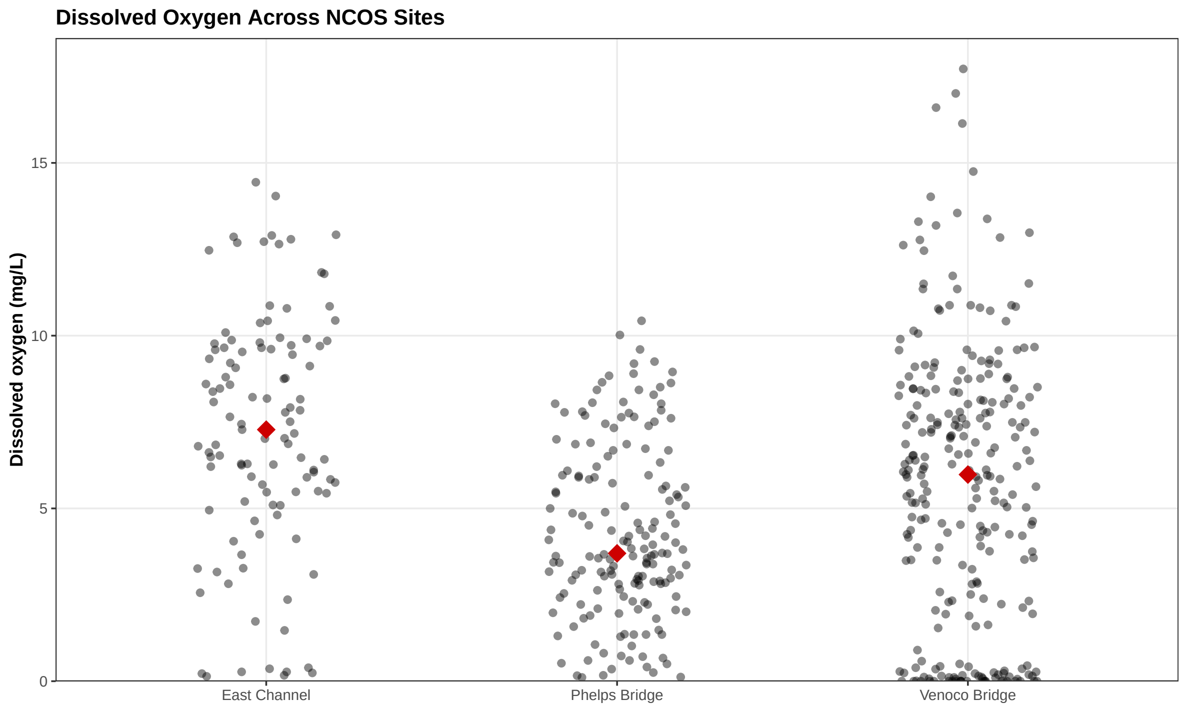
```
# A tibble: 9 x 6
  site_full     elevation_code mean_do median_do sd_do     n
  <chr>         <fct>            <dbl>     <dbl> <dbl> <int>
1 East Channel  0                 7.02      6.66  3.62    54
2 East Channel  1                 7.53      7.96  3.20    46
3 East Channel  2                 6.86      7.17  3.64     9
4 Phelps Bridge 0                 4.53      4.14  2.34    76
5 Phelps Bridge 1                 4.19      3.63  2.61    81
6 Phelps Bridge 2                 3.31      2.01  3.08     9
7 Venoco Bridge 0                 4.01      4.31  3.40    89
8 Venoco Bridge 1                 6         6.38  4.16    99
9 Venoco Bridge 2                 7.38      7.62  3.72    69
```

6. Visualizing

These figures describe patterns in dissolved oxygen before the statistical tests. They compare DO across sites, across elevation-code measurement positions, and with water temperature. The figures are descriptive because the sites also differ in tidal influence, salinity, flow, and distance from water sources.

Figure 1: Dissolved oxygen across sites

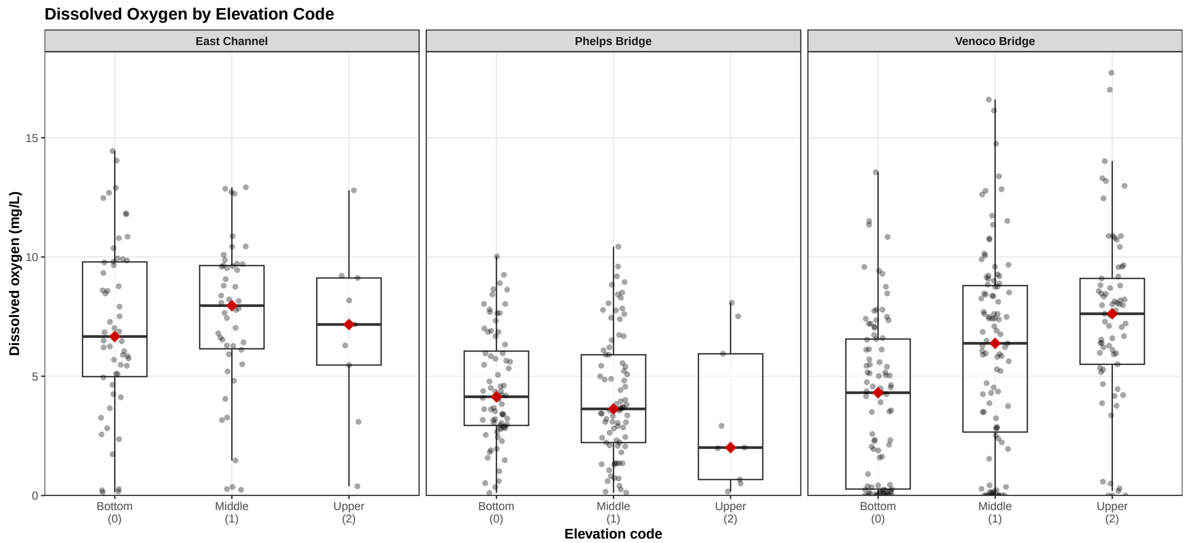
This figure compares dissolved oxygen across the three main NCOS sites. Points show individual observations, and red diamonds show site medians.



Dissolved oxygen varies by site. East Channel has the highest median DO, and Phelps Bridge has the lowest. Venoco Bridge has a median between the other two sites, but it has the widest spread, with many values near 0 mg/L and several high DO values.

Figure 2: Dissolved oxygen by elevation code

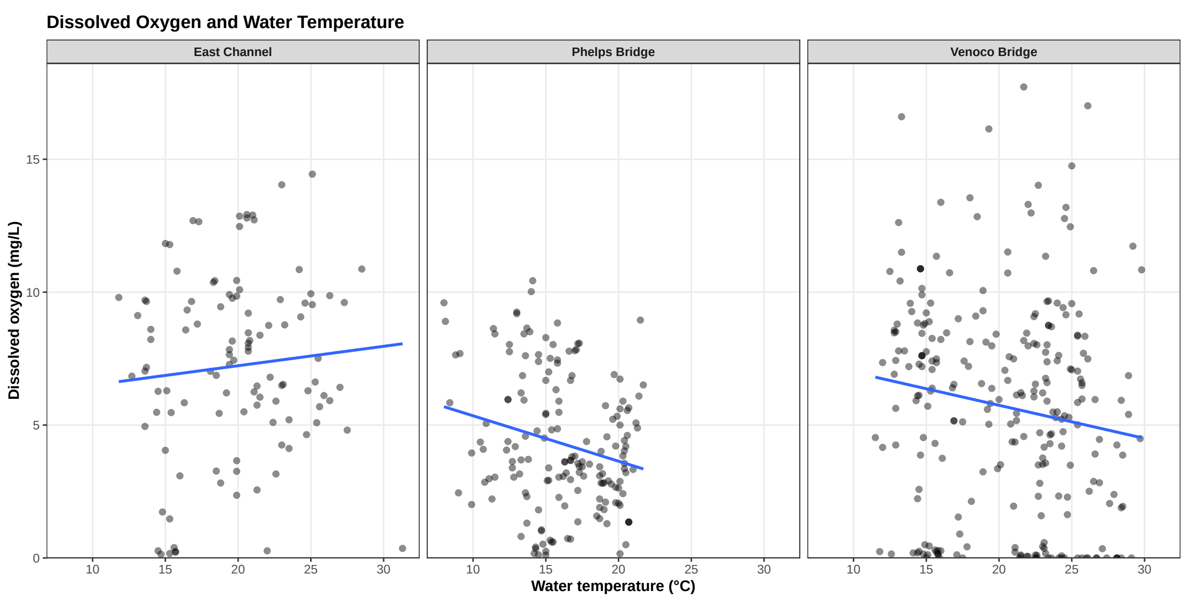
This figure compares dissolved oxygen across elevation-code measurement positions within each site.



The DO pattern across elevation codes is different for each site. East Channel looks similar across the three positions. Phelps Bridge stays low across positions, with fewer observations at elevation code 2. Venoco Bridge has lower DO at elevation code 0 and higher DO at elevation codes 1 and 2.

Figure 3: Dissolved oxygen and water temperature

This figure shows dissolved oxygen and water temperature within each site. The fitted lines summarize the visual pattern in each panel.



The temperature-DO pattern is not the same across sites. East Channel has a slight upward fitted line, while Phelps Bridge and Venoco Bridge have downward fitted lines. The points are widely scattered, especially at Venoco Bridge, where low and high DO values occur across similar temperatures.

Figure 4: Dissolved oxygen, temperature, site, and elevation code

This figure separates the temperature-DO plot by site and elevation code.

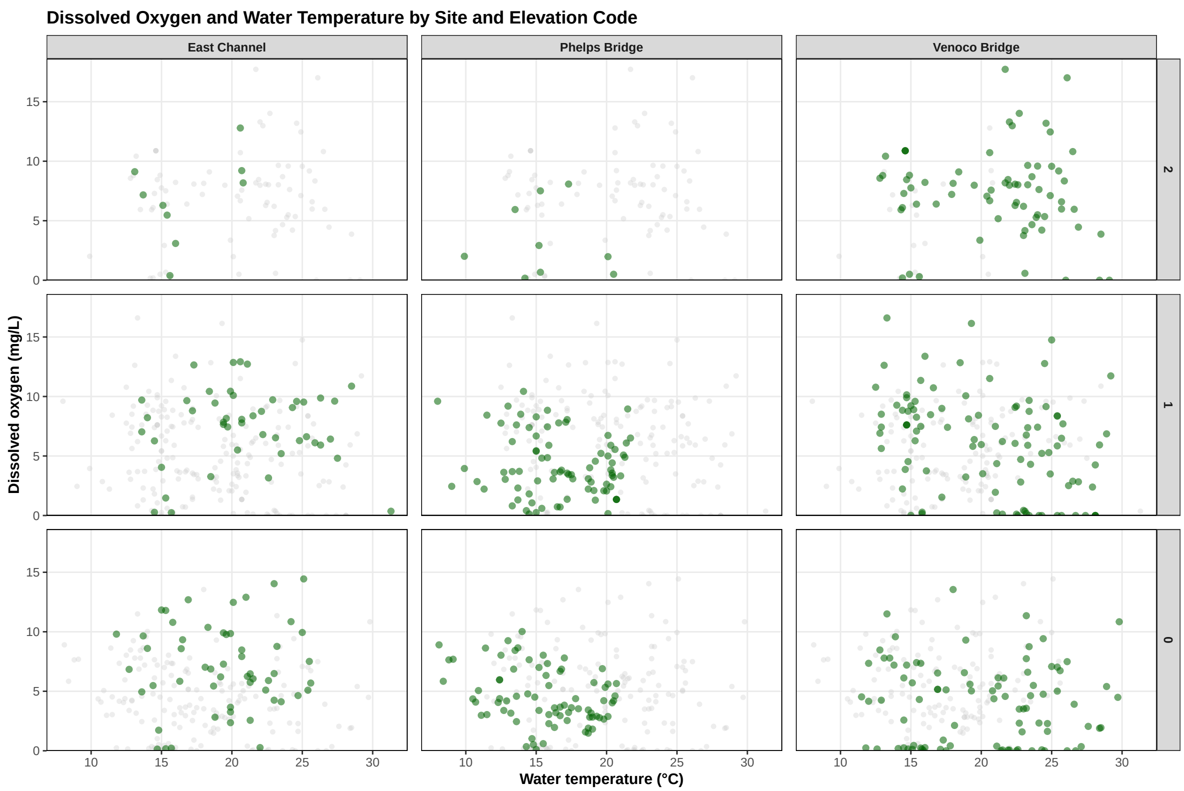
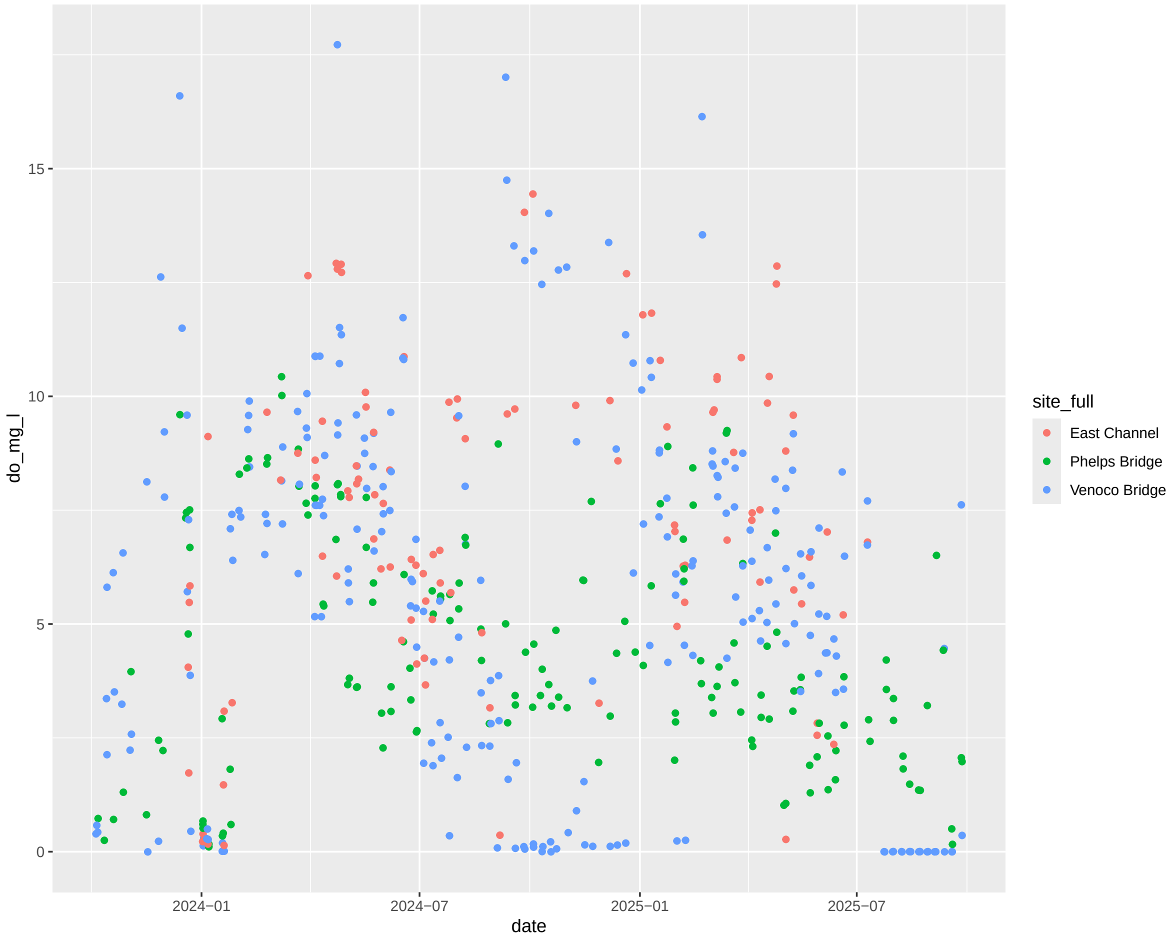
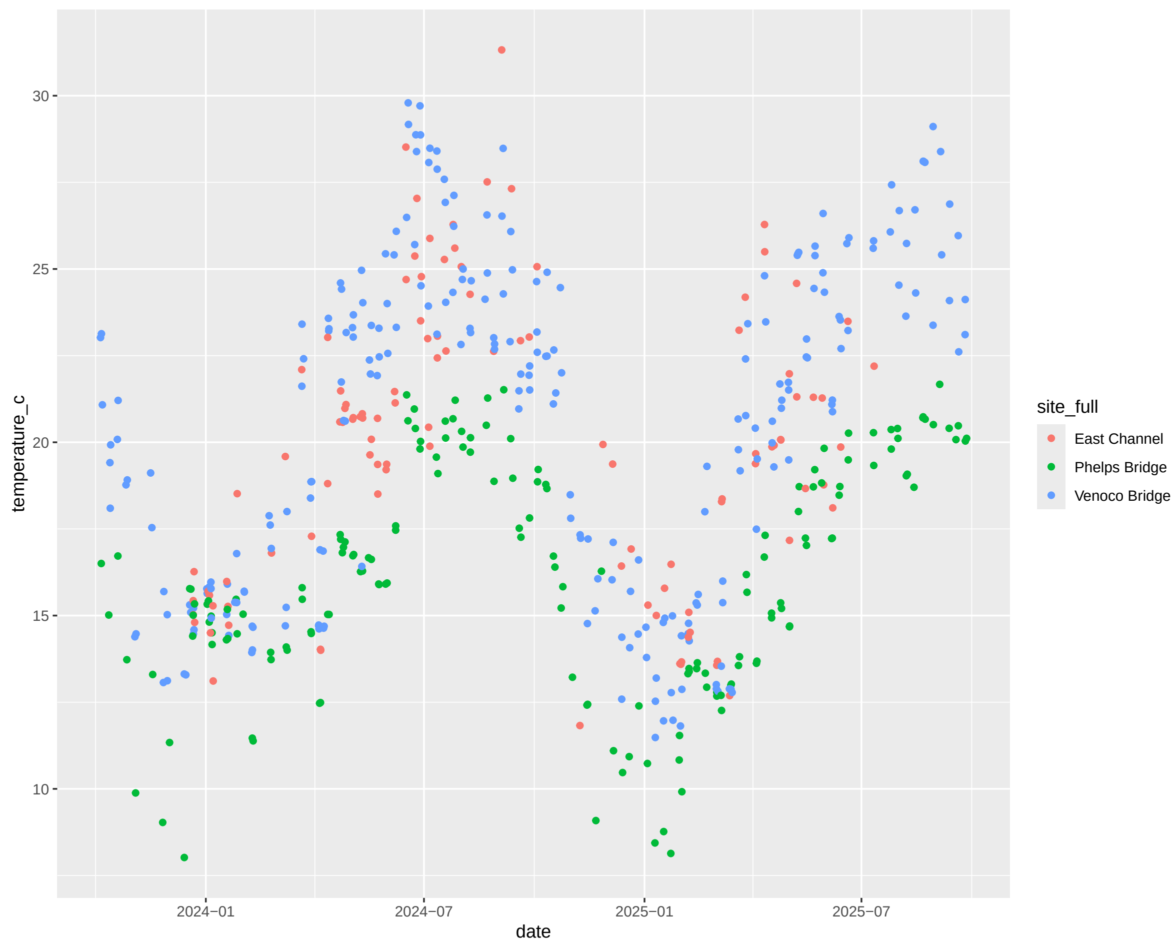


Figure 4 shows DO and temperature separated by site and elevation code. Venoco Bridge has the clearest difference across elevation codes, with lower DO at elevation code 0 and higher DO at elevation codes 1 and 2. Phelps Bridge stays relatively low across positions, and East Channel looks more even across positions. DO values do not separate clearly by temperature.

Figures 5, 6: Time series exploration





7. Statistical analysis

The statistical analyses test whether dissolved oxygen differs across sites, whether dissolved oxygen differs across elevation codes, and whether dissolved oxygen is related to temperature. Kruskal-Wallis tests are used for group comparisons with more than two groups. Pairwise Wilcoxon tests are used after significant Kruskal-Wallis results. Linear models are used for the temperature analyses.

Dissolved oxygen across sites

The first analysis tests whether dissolved oxygen differs across East Channel, Phelps Bridge, and Venoco Bridge.

```
	Kruskal-Wallis rank sum test

data:  do_mg_l by site_full
Kruskal-Wallis chi-squared = 45.667, df = 2, p-value = 1.212e-10
```

Dissolved oxygen differs significantly across sites (Kruskal-Wallis test, $\chi^2 = 45.67$, df = 2, $p < 0.001$). This means that East Channel, Phelps Bridge, and Venoco Bridge do not have the same dissolved oxygen distribution.

```
	Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data:  do_filtered$do_mg_l and do_filtered$site_full

              East Channel Phelps Bridge
Phelps Bridge 3.1e-12      -
Venoco Bridge 0.00026      0.00044

P value adjustment method: BH
```

Pairwise Wilcoxon tests show that all three sites differ from each other. Phelps Bridge differs from East Channel ($p < 0.001$), Venoco Bridge differs from East Channel ($p < 0.001$), and Venoco Bridge differs from Phelps Bridge ($p < 0.001$). East Channel has the highest median DO, Phelps Bridge has the lowest median DO, and Venoco Bridge has the greatest spread.

Dissolved oxygen across elevation codes

The second analysis tests whether dissolved oxygen differs across elevation codes 0, 1, and 2 across the full filtered dataset. Elevation codes represent measurement positions in the water column.

```
	Kruskal-Wallis rank sum test

data:  do_mg_l by elevation_code
Kruskal-Wallis chi-squared = 18.997, df = 2, p-value = 7.498e-05
```

Dissolved oxygen differs significantly across elevation codes overall (Kruskal-Wallis test, $\chi^2 = 19.00$, df = 2, $p < 0.001$). This means that dissolved oxygen is not distributed the same way across bottom, middle, and upper measurements when all sites are analyzed together.

```
    Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data:  do_filtered$do_mg_l and do_filtered$elevation_code

  0       1
1 0.032   -
2 2.7e-05 0.016

P value adjustment method: BH
```

Pairwise Wilcoxon tests show significant differences between all elevation code pairs overall. Elevation code 0 differs from elevation code 1 ($p = 0.032$), elevation code 0 differs from elevation code 2 ($p < 0.001$), and elevation code 1 differs from elevation code 2 ($p = 0.016$).

Dissolved oxygen across elevation codes within each site

The next analysis compares dissolved oxygen across elevation codes within each site. This checks whether the elevation-code pattern is the same at each location.

```
    Kruskal-Wallis rank sum test

data:  do_mg_l by elevation_code
Kruskal-Wallis chi-squared = 1.1119, df = 2, p-value = 0.5735


    Kruskal-Wallis rank sum test

data:  do_mg_l by elevation_code
Kruskal-Wallis chi-squared = 3.0201, df = 2, p-value = 0.2209


    Kruskal-Wallis rank sum test

data:  do_mg_l by elevation_code
Kruskal-Wallis chi-squared = 28.641, df = 2, p-value = 6.034e-07
```

Dissolved oxygen does not differ significantly across elevation codes at East Channel ($p = 0.574$) or Phelps Bridge ($p = 0.221$). At Venoco Bridge, dissolved oxygen differs significantly across elevation codes (Kruskal-Wallis test, $\chi^2 = 28.64$, df = 2, $p < 0.001$). This shows that the overall elevation-code result is mostly coming from the Venoco Bridge pattern.

```
    Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data:  pull(filter(do_filtered, site_full == "Venoco Bridge"), do_mg_l) and pull(filter(do_fi

  0       1
1 0.0018  -
2 1.6e-07 0.0417

P value adjustment method: BH
```

At Venoco Bridge, dissolved oxygen differs significantly between all elevation code pairs. Elevation code 0 differs from elevation code 1 ($p = 0.0018$), elevation code 0 differs from elevation code 2 ($p < 0.001$), and elevation code 1 differs from elevation code 2 ($p = 0.0417$). Median dissolved oxygen increases from elevation code 0 to elevation code 2: 4.31 mg/L at code 0, 6.38 mg/L at code 1, and 7.62 mg/L at code 2. This matches the visual pattern in Figure 2, where Venoco Bridge shows the clearest difference across elevation codes.

Dissolved oxygen and temperature

Temperature is included as an exploratory variable to test whether water temperature is related to dissolved oxygen.

```
Call:
lm(formula = do_mg_l ~ temperature_c, data = do_filtered)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.5923 -2.6991  0.0171  2.5303 12.1780 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)    5.73733    0.66236   8.662   <2e-16 ***
temperature_c -0.00900    0.03383  -0.266     0.79    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.635 on 530 degrees of freedom
Multiple R-squared:  0.0001335, Adjusted R-squared:  -0.001753 
F-statistic: 0.07078 on 1 and 530 DF,  p-value: 0.7903
```

Water temperature alone is not significantly related to dissolved oxygen (linear model, p = 0.790, $R^2 < 0.001$). Temperature by itself does not describe the DO pattern in the full filtered dataset.

Dissolved oxygen, site, elevation code, and temperature

A combined model tests whether dissolved oxygen is related to site, elevation code, and temperature together.

```
Call:
lm(formula = do_mg_l ~ site_full + elevation_code + temperature_c,
    data = do_filtered)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.7848 -2.1683  0.0152  2.2950 11.0203 

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)    
(Intercept)            8.78887    0.77032  11.409  < 2e-16 ***
site_fullPhelps Bridge -3.32780    0.44018  -7.560 1.80e-13 ***
site_fullVenoco Bridge -1.82445    0.39730  -4.592 5.49e-06 ***
elevation_code1        0.91901    0.32455   2.832  0.00481 ** 
elevation_code2        2.04989    0.44770   4.579 5.84e-06 ***
temperature_c         -0.10667    0.03501  -3.047  0.00243 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.402 on 526 degrees of freedom
Multiple R-squared:  0.1308,	Adjusted R-squared:  0.1226 
F-statistic: 15.84 on 5 and 526 DF,  p-value: 1.532e-14
```

In the combined model, site, elevation code, and temperature are significant. Phelps Bridge and Venoco Bridge differ from East Channel, elevation codes 1 and 2 differ from elevation code 0, and temperature has a negative relationship with dissolved oxygen after site and elevation code are included.

The combined model explains about 13% of the variation in dissolved oxygen ($R^2 = 0.131$). This means these variables are related to dissolved oxygen, but most of the variation is still not explained by this model.

Statistical interpretation

Dissolved oxygen differs significantly across sites. Pairwise tests show that all three sites differ from each other. East Channel has the highest median DO, Phelps Bridge has the lowest median DO, and Venoco Bridge has the greatest spread.

Dissolved oxygen also differs significantly across elevation codes overall. However, the within-site tests show that this pattern is mostly from Venoco Bridge. East Channel and Phelps Bridge do not show significant DO differences across elevation codes, while Venoco Bridge does. At Venoco Bridge, median DO increases from elevation code 0 to elevation code 2.

Temperature alone is not significantly related to dissolved oxygen. Temperature is significant in the combined model after site and elevation code are included, but the model only explains about 13% of the variation.

Overall, the strongest results are the site differences and the elevation-code pattern at Venoco Bridge. Temperature alone is not significant, but it is significant when site and elevation code are included. Future analysis could include salinity, flow, tidal timing, and distance from water sources to better describe the site-level differences.